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Inventor(s)	Newbold; Tyler L.

Armrest control mechanism for passenger seat

Abstract

An armrest control mechanism for a passenger seat may be attached to a seat frame and an armrest to control movement of the armrest relative to the seat frame. The armrest control mechanism may include a base member, a gas spring, a pulley attached to the gas spring and moveable relative to the base member, and a cable movable relative to the base member for moving the pulley. The gas spring may provide forces through the pulley and the cable to the armrest.

Inventors: **Newbold; Tyler L. (Sherman, TX)**

Applicant: **Safran Seats USA LLC** (Gainesville, TX)

Family ID: **92909516**

Assignee: **Safran Seats USA LLC (Gainesville, TX)**

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Primary Examiner: Gabler; Philip F

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

FIELD OF THE INVENTION

(1) The field of the invention relates to passenger seats and, more particularly, to systems and methods for controlling movement of armrests from a stowed position to a deployed position.

BACKGROUND

(2) Passenger vehicles, such as aircraft, busses, trains, ships, and automobiles often include one or more passenger seats in which passengers may be seated and otherwise use during travel. Many passenger seats include one or more armrests that a passenger may use to rest his or her arm. Some armrests are pivotable relative to the passenger seat, and movement between a stowed position and a deployed position is achieved by pivoting the armrest about a pivot axis. Other armrests are translatable or movable in a vertical direction between a stowed (or high) position in which the passenger can comfortably rest his or her arm and a deployed (or low) position in which the armrest may be the same height as a base or seating part of the passenger seat. In some cases, vertically movable armrests may be movable between the stowed and deployed positions so as to facilitate access to the seat for a disabled person. Traditional approaches to controlling movement of vertically movable armrests have relied upon the strength of the flight attendant and/or coil springs. For example, a flight attendant may be required to support the entire weight of the armrest to avoid free fall of the armrest when moving from the stowed position to the deployed position and conversely may be required to lift the entire weight when moving from the deployed position to the stowed position. Coil springs may produce insufficient forces to support or control movement of the armrest and/or may result in undesired oscillations of the armrest.

SUMMARY

(3) The terms “invention,” “the invention,” “this invention” and “the present invention” used in this patent are intended to refer broadly to all of the subject matter of this patent and the patent claims

below. Statements containing these terms should be understood not to limit the subject matter described herein or to limit the meaning or scope of the patent claims below. Embodiments of the invention covered by this patent are defined by the claims below, not this summary. This summary is a high-level overview of various aspects of the invention and introduces some of the concepts that are further described in the Detailed Description section below. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this patent, any or all drawings and each claim.
(4) According to certain embodiments of the present disclosure, a passenger seat includes a seat frame, an armrest moveable relative to the seat frame, and an armrest control mechanism attached to the seat frame and the armrest. The armrest control mechanism includes a base member, a gas spring, a pulley, and a cable. The gas spring includes a first end attached to the base member and a second end moveable relative to the base member, and the pulley is attached to the second end of the gas spring and moveable relative to the base member. The cable includes a first end, a second end, and an intermediate portion. The first end of the cable is attached to an anchor location on the base member, the second end of the cable is moveable relative to the base member, and the intermediate portion extends at least partially around the first pulley and engages with the first pulley. In various embodiments, the first pulley is moveable based on a movement of the second end of the cable. In certain embodiments, the base member is attached to one of the seat frame or the armrest and the second end of the cable is attached to the other of the seat frame or the armrest.
(5) According to certain embodiments of the present invention, a method of assembling an armrest for a passenger seat relative to a seat frame of the passenger seat includes supporting the armrest relative to the seat frame such that the armrest is vertically movable between a stowed position and a deployed position, and attaching an armrest control mechanism to the seat frame and the armrest for controlling vertical movement of the armrest relative to the seat frame.
(6) According to certain embodiments of the present invention, an armrest control mechanism includes a base member, a gas spring, a pulley, and a cable. The gas spring includes a piston and cylinder, where the piston is movable relative to the cylinder and the cylinder is attached to the base member. The pulley is attached to the piston of the gas spring such that the first pulley is moveable relative to the base member. The cable includes a first end, a second end, and an intermediate portion between the first end and the second end. In various embodiments, the first end of the cable is attached to the base member, the second end of the cable is moveable relative to the base member, and the intermediate portion extends at least partially around and engages with the first pulley. In certain embodiments, the first pulley is moveable based on a movement of the second end of the cable. In various embodiments, the base member is attachable to one of a seat frame of a passenger seat or an armrest of the passenger seat and the second end of the cable is attachable to the other of the seat frame or the armrest.
(7) Various implementations described herein can include additional systems, methods, features, and advantages, which cannot necessarily be expressly disclosed herein but will be apparent to one of ordinary skill in the art upon examination of the following detailed description and accompanying drawings. It is intended that all such systems, methods, features, and advantages be included within the present disclosure and protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The specification makes reference to the following appended figures, in which use of like reference numerals in different figures is intended to illustrate like or analogous components.

(2) FIG. **1** is a perspective view of a passenger seat assembly with an armrest and an armrest control mechanism according to embodiments.
(3) FIG. **2**A illustrates the armrest of FIG. **1** in a deployed position.
(4) FIG. **2**B illustrates the armrest of FIG. **1** in a stowed position.
(5) FIGS. **3**A-B are top perspective views of a portion of the armrest control mechanism of FIG. **1** in a stowed configuration.
(6) FIGS. **4**A-B are top perspective views of the portion of the armrest control mechanism of FIG. **1** in a deployed configuration.
(7) FIG. **5** is a bottom perspective view of the portion of the armrest control mechanism of FIG. **1**.
(8) FIG. **6** is another bottom perspective view of the portion of the armrest control mechanism of FIG. **1** with a bottom cover removed.
(9) FIG. **7** is a perspective view of an armrest control mechanism according to embodiments.
(10) FIG. **8** is another perspective view of the armrest control mechanism of FIG. **7**.
(11) FIG. **9** illustrates a portion of the armrest control mechanism of FIG. **7** taken from box **9** in FIG. **8**.
(12) FIG. **10** illustrates the armrest control mechanism of FIG. **7** with a cover according to embodiments.

DETAILED DESCRIPTION

(13) Described herein are systems and methods for controlling vertical movement of an armrest for passenger seats. While the armrest control mechanisms are discussed for use with aircraft seats, they are by no means so limited. Rather, embodiments of the armrest control mechanism may be used in passenger seats or other seats of any type or otherwise as desired.
(14) FIGS. **1-2**B illustrate a passenger seat assembly **100** according to various embodiments. The passenger seat assembly **100** is illustrated with two passenger seats **102**A-B, although in other embodiments, the passenger seat assembly **100** may include any number of passenger seats **102** as desired, such as but not limited to a single passenger seat **102**, two passenger seats **102**, three passenger seats **102**, etc. Each passenger seat **102** generally includes a seat frame **104** for supporting one or more cushioning and/or support elements for supporting a passenger such as but not limited to a seat base **108** and/or a backrest **110**. In embodiments with a plurality of seats **102**, a common seat frame component (e.g., a frame for the seat base **108**) optionally may be used for more than one seat **102**, although it need not in other embodiments.
(15) In various embodiments, and as illustrated in FIG. **1**, the passenger seat assembly **100** includes one or more armrests **106** associated with a given passenger seat **102** for a passenger to rest his or her arm. Whereas other passenger seats may utilize pivotable armrests that are pivotable between a stowed position and deployed position, the armrests **106** described herein are vertically movable or translatable between a stowed (or high) position and a deployed (or low) position. In FIG. **1**, the armrest **106**A is in the deployed position and armrests **106**B-C are in the stowed positions. Similarly, FIG. **2**A illustrates the armrest **106**A in the stowed position and FIG. **2**B illustrates the armrest **106**B in the deployed position. As illustrated, in the deployed position, the armrest **106** may be at a lowered position relative to a seat base **108**. In certain embodiments, the armrest **106** may be movable to the deployed position to facilitate access to the passenger seat **102** for a disabled person. Various support mechanisms may be utilized to support the armrest **106** relative to the seat frame **104** such that the armrest **106** is vertically movable, such as but not limited to rails, linear actuators, links, hinges, combinations thereof, and/or other mechanisms or combinations of mechanisms as desired.
(16) In various embodiments, the passenger seat assembly **100** includes an armrest control mechanism **112** for facilitating control of movement of the armrest **106** relative to the seat frame **104** between the stowed and deployed positions. The armrest control mechanism **112** may be attached to both the seat frame **104** and the armrest **106**. In some embodiments, the armrest control mechanism **112** may resist motion of the armrest **106** moving downwards from the stowed position

to the deployed position and may prevent the armrest **106** from free-falling downwards. Additionally, or alternatively, the armrest control mechanism **112** may provide an assist load for moving the armrest **106** upwards from the deployed position to the stowed position. Compared to traditional approaches, the armrest control mechanism **112** may provide a compact mechanism for reliably controlling vertical movement of the armrest **106**. In one non-limiting example, the armrest control mechanism **112** advantageously may provide a reliable way to control deployment speed of a high-weight armrest while fitting within a small space, such as a space defined within the armrest and/or on the seat frame **104**. Various other benefits and advantages may be realized with the systems and methods described herein, and the aforementioned benefits and advantages should not be considered limiting.

(17) FIGS. **3**A-**6** illustrate the armrest control mechanism **112** in greater detail. In various embodiments, the armrest control mechanism **112** includes a base member **114**, a gas spring **116**, a cable **118**, a first pulley **120**, and a second pulley **122**.

(18) The base member **114** generally includes a first side **124** and a second side **125** opposite from the first side **124**. In some embodiments, the base member **114** may be supported on the armrest **106**, while in other embodiments, the base member **114** may be supported on the seat frame **104**.

(19) The gas spring **116** includes a cylinder **126** and a piston **128** that is movable relative to the cylinder **126**. In certain embodiments, the piston **128** is linearly movable relative to the cylinder **126** between an extended position (FIGS. **3**A-B) and a retracted position (FIGS. **4**A-B). In various embodiments, and as discussed in detail below, the extended position of the piston **128** may correspond with the stowed position of the armrest **106** and the retracted position of the piston **128** may correspond with the deployed position of the armrest **106**. In various embodiments, the cylinder **126** may define a first end **130** of the gas spring **116** and the piston **128** defines a second end **132** of the gas spring **116**. As best illustrated in FIGS. **3**A and **4**A, the gas spring **116** may be supported on one of the sides (e.g., the first side **124**) of the base member **114** using various mechanisms as desired, such as but not limited to mechanical fasteners. In certain embodiments, the gas spring **116** is supported on the base member **114** such that the first end **130** is fixed relative to the base member **114** while the second end **132** is movable relative to the base member **114**.

(20) Optionally, and as illustrated in FIGS. **3**A-**4**B, the base member **114** may define a guide track **148**. In such embodiments, a guide member **150** may be supported on the piston **128**, optionally at or proximate to the second end **132**, and movable along the guide track **148**. The guide member **150** and guide track **148** may facilitate movement of the piston **128** between the extended position and the retracted position.

(21) The cable **118** of the armrest control mechanism **112** generally includes a first end **134**, a second end **136**, and an intermediate portion **138** between the first end **134** and the second end **136**. In various embodiments, the first end **134** of the cable **118** may be attached to an anchor location **140** on the base member **114** such that the first end **134** is fixed relative to the base member **114**. The second end **136** of the cable **118** may be attached to the seat frame **104** or the armrest **106**. In various embodiments, the base member **114** may be supported on one of the seat frame **104** or the armrest **106**, and the second end **136** of the cable **118** may be attached to the other of the seat frame **104** or the armrest **106**. In the embodiment best illustrated in FIGS. **2**A-B, the base member **114** is supported on the armrest **106** and the second end **136** attached to the seat frame **104**.

(22) Attachment of the second end **136** of the cable **118** to the armrest **106** and/or the seat frame **104** allows for relative movement between the second end **136** and the base member **114**, which in turn facilitates movement of the armrest **106** between the stowed position and the deployed position. In various embodiments, the second end **136** is movable between a stowed position (see FIGS. **3**A-B) and a deployed position (see FIGS. **4**A-B) relative to the base member **114**. The stowed position of the second end **136** may correspond with the stowed position of the armrest **106** and the deployed position of the second end **136** may correspond with the deployed position of the armrest **106**.

(23) The first pulley **120** and the second pulley **122** are adapted to receive and guide the cable **118** on the base member **114**. In certain embodiments, the first pulley **120** is supported on the piston **128**, optionally at or proximate to the second end **132**, such that the first pulley **120** is movable relative to the base member **114**. The second pulley **122** of the armrest control mechanism **112** may be supported on the base member **114** at a fixed location. In some embodiments, the fixed location of the second pulley **122** is at a vertical position below the anchor location **140**, although in other embodiments, in need not be. As a non-limiting example, the fixed location may be at a vertical position above the anchor location **140**. In some embodiments, the fixed location and anchor location **140** are vertically aligned, although they need not be in other embodiments.
(24) The first pulley **120** and the second pulley **122** together define a guide path for the cable **118**. In some embodiments, the guide path for the cable **118** may be defined on one side (e.g., the first side **124**) of the base member **114**, while in other embodiments and as illustrated in FIGS. **3**A-**6**, the guide path for the cable **118** may be on both sides **124**, **125** of the base member **114**. As such, depending on the desired guide path, the armrest control mechanism **112** optionally includes supplemental pulleys in addition to the first pulley **120** and the second pulley **122**. In the embodiment illustrated in FIGS. **3**A-**6**, the armrest control mechanism **112** further includes an optional crossover pulley **146** for guiding the cable **118** from the second pulley **122** on the first side **124** to the second side **125** of the base member **114**.
(25) As best illustrated in FIGS. **3**A-**4**B, in certain embodiments, the guide path for the cable **118** extends from the anchor location **140** to the first pulley **120**, from the first pulley **120** to the second pulley **122**, and from the second pulley **122** towards the armrest **106** and/or seat frame **104** to which the second end **136** of the cable **118** is attached. In such embodiments, the intermediate portion **138** of the cable **118** may extend at least partially around the first pulley **120** and at least partially around the second pulley **122**.
(26) In various embodiments, the guide path for the cable **118** from the anchor location **140** to the first pulley **120** may extend in a first direction and the guide path for the cable **118** from the second pulley **122** towards the armrest **106** and/or the seat frame **104** may be in a second direction that is different from the first direction. In various embodiments, the second direction may be perpendicular to the second direction, although it need not be in other embodiments. As a non-limiting example, when the armrest control mechanism **112** is installed with passenger seat **102**, the first direction may be in a forward-aft direction, and the second direction may be in a side-to-side direction and/or a vertical direction. In a non-limiting example, the first direction may be the forward-aft direction, and the second direction may be in the vertical direction.
(27) Optionally, the guide path from the second pulley **122** towards the armrest **106** and/or the seat frame **104** may be in a plurality of directions (e.g., the second direction and at least one other direction) that is different from the first direction, such as but not limited to when the armrest control mechanism **112** includes the crossover pulley **146**. In the embodiment illustrated in FIGS. **3**A-**6**, the first direction is in the forward-aft direction, and the guide path from the second pulley **122** towards the armrest **106** extends both in a vertical direction and a side-to-side direction via the crossover pulley **146**. Various other configurations of the guide path may be utilized as desired.
(28) In various embodiments, the guide path for the cable **118** from the first pulley **120** to the second pulley **122** may be in the same direction and/or in a direction that is parallel to that of the first direction. In certain embodiments, the first direction may be parallel to a direction of linear movement of the piston **128** of the gas spring **116**.
(29) In certain embodiments, a length of travel of the second end **136** between the stowed position (see FIGS. **3**A-B) and the deployed position (see FIGS. **4**A-B) may directly correlate with a length of travel of the first pulley **120** (via the piston **128**) relative to the anchor location **140** (e.g., between retracted and extended positions of the piston **128**). In certain embodiments, the length of travel of the second end **136** between the stowed and deployed positions may be twice (2×) the length of travel of the first pulley **120**.

(30) Optionally, the cable **118** includes one or more stoppers **142** along the cable **118** for controlling a length of travel of the cable **118** relative to the base member **114** and/or for defining the stowed position of the cable **118** relative to the base member **114**. When the one or more stoppers **142** are included, the stoppers **142** may engage a corresponding stopper **152** on the base member **114** to limit further relative movement of the cable **118**. In some embodiments, and as best illustrated in FIGS. **5** and **6**, the stopper **152** may be a stopper plate **156** supported on the base **114**. FIGS. **7-10** illustrate another example of an armrest control system **712** that is similar to the armrest control system **712** except that the stopper **152** is provided directly on the crossover pulley **146**. Other configurations of stoppers may be utilized as desired. Moreover, as previously discussed, the crossover pulley **146** need not be included, and the guide path for the cable **118** optionally may be provided only on a single side of the base member **114**.
(31) Optionally, and as illustrated in FIG. **10**, one or more covers **758** may be provided to house and/or substantially cover the components of the armrest control systems described herein. When included, the one or more covers **758** may isolate the components of the armrest control system from other components of the armrest **106** and/or seat frame **104** and/or may provide one or more locations for mounting the armrest control system on the armrest **106** and/or seat frame **104**.
(32) Referring back to FIGS. **2**A-B, the armrest control mechanism **112** may assist in raising and lowering the armrest **106** by applying forces through a cable **118** and using the gas spring **116**. In various examples, the armrest control mechanism **112**, using the gas spring **116**, may apply a resistive force to the armrest **106** as the armrest **106** moves in a downward direction, towards the deployed position illustrated in FIG. **2**A. In particular, a rate at which the piston **128** moves from the extended position to the retracted position may provide a force to the cable **118** resisting and/or counteracting the weight of the armrest **106**, thereby providing a reliable and controlled movement of the armrest **106** from the stowed position to the deployed position. Conversely, the armrest control mechanism **112**, using the gas spring **116**, may apply an assisting force to the armrest **106** as the armrest **106** moves in an upward direction, towards the stowed position illustrated in FIG. **2**B. In particular, the piston **128** may be driven and/or otherwise forced to move from the retracted position to the extended position, thereby applying a force to the cable **118** assisting with the upward movement of the armrest **106**, which provides a reliable and controlled movement of the armrest **106** from the deployed position to the stowed position.
(33) Optionally, and as illustrated in FIGS. **2**A-B, for example, the armrest control mechanism **112** includes a latch **160**, which is movable between an engaged state and a disengaged state. In the engaged state, latch **160** may restrict and/or prevent movement of the armrest **106** relative to the seat frame **104**, while in the disengaged state, the armrest **106** may be movable relative to the seat frame **104**.
(34) A collection of exemplary embodiments is provided below, including at least some explicitly enumerated as an “Illustration” providing additional description of a variety of example embodiments in accordance with the concepts described herein. These illustrations are not meant to be mutually exclusive, exhaustive, or restrictive; and the disclosure not limited to these example illustrations but rather encompasses all possible modifications and variations within the scope of the issued claims and their equivalents.
(35) Illustration 1. A passenger seat comprising: a seat frame; an armrest moveable relative to the seat frame; and an armrest control mechanism attached to the seat frame and the armrest, the armrest control mechanism comprising: a base member; a gas spring comprising a first end attached to the base member and a second end moveable relative to the base member; a first pulley attached to the second end of the gas spring and moveable relative to the base member; and a cable comprising a first end, a second end, and an intermediate portion, wherein the first end of the cable is attached to an anchor location on the base member, the second end of the cable is moveable relative to the base member, and the intermediate portion extends at least partially around the first pulley and engages with the first pulley, wherein the first pulley is moveable based on a movement

of the second end of the cable, and wherein the base member is attached to one of the seat frame or the armrest and the second end of the cable is attached to the other of the seat frame or the armrest.

(36) Illustration 2. The passenger seat of any preceding or subsequent illustration or combination of illustrations, wherein the base member comprises a guide track and the gas spring further comprises a guide member at the second end, wherein the first pulley is attached to the guide member, and wherein the guide member is movable along the guide track.

(37) Illustration 3. The passenger seat of any preceding or subsequent illustration or combination of illustrations, wherein the cable extends in a first direction from the first end to the first pulley and in a second direction from the base member to the second end, wherein the second direction is different from the first direction.

(38) Illustration 4. The passenger seat of any preceding or subsequent illustration or combination of illustrations, wherein the first direction is perpendicular to the second direction.

(39) Illustration 5. The passenger seat of any preceding or subsequent illustration or combination of illustrations, wherein the second end of the cable is movable between a stowed position and a deployed position, and wherein the first pulley is linearly movable between a stowed position and a deployed position, and wherein a length of travel of the second end of the cable between the stowed position and the deployed position is different from a length of travel of the first pulley from the stowed position to the deployed position.

(40) Illustration 6. The passenger seat of any preceding or subsequent illustration or combination of illustrations, wherein the length of travel of the second end between the stowed position and the deployed position is twice the length of travel of the first pulley between the stowed position and the deployed position.

(41) Illustration 7. The passenger seat of any preceding or subsequent illustration or combination of illustrations, wherein the armrest control mechanism applies a resistive force to the armrest as the armrest moves in a downward direction, and the armrest control mechanism applies an assisting force to the armrest as the armrest moves in an upward direction.

(42) Illustration 8. The passenger seat of any preceding or subsequent illustration or combination of illustrations, further comprising a release latch attached to the armrest, wherein the release latch is operable between an engaged state and a disengaged state, wherein, in the engaged state, the release latch locks the armrest relative to the seat frame, and wherein, in the disengaged state, the armrest is moveable relative to the seat frame.

(43) Illustration 9. A method of assembling an armrest for a passenger seat relative to a seat frame of the passenger seat, the method comprising: supporting the armrest relative to the seat frame such that the armrest is vertically movable between a stowed position and a deployed position; and attaching an armrest control mechanism to the seat frame and the armrest for controlling vertical movement of the armrest relative to the seat frame, wherein the armrest control mechanism comprises: a base member; a gas spring comprising a first end attached to the base member and a second end moveable relative to the first end and relative to the base member; a first pulley attached to the second end of the gas spring and moveable relative to the base member; and a cable comprising a first end, a second end, and an intermediate portion between the first end and the second end, wherein the first end of the cable is attached to an anchor location on the base member, the second end of the cable is moveable relative to the base member, and the intermediate portion extends at least partially around and engages with the first pulley, wherein the first pulley is moveable based on a movement of the second end of the cable, and wherein attaching the armrest control mechanism to the seat frame and the armrest comprises attaching the base member to one of the seat frame or the armrest and attaching the second end of the cable to the other of the seat frame or the armrest.

(44) Illustration 10. The method of any preceding or subsequent illustration or combination of illustrations, wherein attaching the armrest control mechanism comprises guiding the cable such that the cable extends in a first direction from the first end to the first pulley and in a second

direction from the base member to the second end, wherein the second direction is different from the first direction.
(45) Illustration 11. The method of any preceding or subsequent illustration or combination of illustrations, wherein the first direction is perpendicular to the second direction.
(46) Illustration 12. The method of any preceding or subsequent illustration or combination of illustrations, wherein the second end of the cable is movable between a stowed position and a deployed position, and wherein the first pulley is linearly movable between a stowed position and a deployed position, and wherein the method comprises guiding the cable on the armrest control mechanism such that a length of travel of the second end of the cable between the stowed position and the deployed position is different from a length of travel of the first pulley from the stowed position to the deployed position, wherein the length of travel of the second end between the stowed position and the deployed position is twice the length of travel of the first pulley between the stowed position and the deployed position.
(47) Illustration 13. An armrest control mechanism comprising: a base member; a gas spring comprising a piston and cylinder, wherein the piston is movable relative to the cylinder, and wherein the cylinder is attached to the base member; a first pulley attached to the piston of the gas spring such that the first pulley is moveable relative to the base member; and a cable comprising a first end, a second end, and an intermediate portion between the first end and the second end, wherein the first end of the cable is attached to the base member, the second end of the cable is moveable relative to the base member, and the intermediate portion extends at least partially around and engages with the first pulley, wherein the first pulley is moveable based on a movement of the second end of the cable, and wherein the base member is attachable to one of a seat frame of a passenger seat or an armrest of the passenger seat and the second end of the cable is attachable to the other of the seat frame or the armrest.
(48) Illustration 14. The armrest control mechanism of any preceding or subsequent illustration or combination of illustrations, wherein the base member comprises a guide track, wherein the first pulley is attached to a guide member, and wherein the guide member is moveable along the guide track.
(49) Illustration 15. The armrest control mechanism of any preceding or subsequent illustration or combination of illustrations, further comprising a second pulley attached to the base member, wherein a guide path for the cable extends from the anchor location, around the first pulley, and around the second pulley toward the seat frame or the armrest.
(50) Illustration 16. The armrest control mechanism of any preceding or subsequent illustration or combination of illustrations, wherein the guide path for the cable between the anchor location and the first pulley is parallel to the guide path for the cable between the anchor location and the second pulley.
(51) Illustration 17. The armrest control mechanism of any preceding or subsequent illustration or combination of illustrations, wherein the guide path for the cable between the anchor location and the first pulley is in a first direction, and wherein a guide path for the cable from the second pulley toward the seat frame or the armrest is in a second direction different from the first direction.
(52) Illustration 18. The armrest control mechanism of any preceding or subsequent illustration or combination of illustrations, wherein the first direction is perpendicular to the second direction.
(53) Illustration 19. The armrest control mechanism of any preceding or subsequent illustration or combination of illustrations, wherein the gas spring and second pulley are supported on a first side of the base member, wherein the armrest further comprises a cable stopper for the cable on a second side of the base member, and wherein the guide path for the cable further extends from the first side of the base member to the second side of the base member.
(54) Illustration 20. The armrest control mechanism of any preceding or subsequent illustration or combination of illustrations, wherein the second end of the cable is movable between a stowed position and a deployed position, and wherein the first pulley is linearly movable between a stowed

position and a deployed position, and wherein the method comprises guiding the cable on the armrest control mechanism such that a length of travel of the second end of the cable between the stowed position and the deployed position is different from a length of travel of the first pulley from the stowed position to the deployed position, wherein the length of travel of the second end between the stowed position and the deployed position is twice the length of travel of the first pulley between the stowed position and the deployed position.

(55) The subject matter of embodiments is described herein with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described. Directional references such as “up,” “down,” “top,” “bottom,” “left,” “right,” “front,” and “back,” among others, are intended to refer to the orientation as illustrated and described in the figure (or figures) to which the components and directions are referencing. In the figures and the description, like numerals are intended to represent like elements. Throughout this disclosure, a reference numeral with a letter refers to a specific instance of an element and the reference numeral without an accompanying letter refers to the element generically or collectively. Thus, as an example (not shown in the drawings), device “12A” refers to an instance of a device class, which may be referred to collectively as devices “12” and any one of which may be referred to generically as a device “12”. As used herein, the meaning of “a,” “an,” and “the” includes singular and plural references unless the context clearly dictates otherwise.

(56) All ranges disclosed herein are to be understood to encompass any and all subranges subsumed therein. For example, a stated range of “1 to 10” should be considered to include any and all subranges between (and inclusive of) the minimum value of 1 and the maximum value of 10; that is, all subranges beginning with a minimum value of 1 or more, e.g., 1 to 6.1, and ending with a maximum value of 10 or less, e.g., 5.5 to 10.

(57) The above-described aspects are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the present disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the present disclosure. All such modifications and variations are intended to be included herein within the scope of the present disclosure, and all possible claims to individual aspects or combinations of elements or steps are intended to be supported by the present disclosure. Moreover, although specific terms are employed herein, as well as in the claims that follow, they are used only in a generic and descriptive sense, and not for the purposes of limiting the described embodiments, nor the claims that follow.

Claims

1. A passenger seat comprising: a seat frame; an armrest moveable relative to the seat frame; and an armrest control mechanism attached to the seat frame and the armrest, the armrest control mechanism comprising: a base member; a gas spring comprising a first end attached to the base member and a second end moveable relative to the base member; a first pulley attached to the second end of the gas spring and moveable relative to the base member; and a cable comprising a first end, a second end, and an intermediate portion, wherein the first end of the cable is attached to an anchor location on the base member, the second end of the cable is moveable relative to the base member, and the intermediate portion extends at least partially around the first pulley and engages with the first pulley, wherein the first pulley is moveable based on a movement of the second end of the cable, and wherein the base member is attached to one of the seat frame or the armrest and the second end of the cable is attached to the other of the seat frame or the armrest.

2. The passenger seat of claim 1, wherein the base member comprises a guide track and the gas spring further comprises a guide member at the second end, wherein the first pulley is attached to the guide member, and wherein the guide member is movable along the guide track.
3. The passenger seat of claim 1, wherein the cable extends in a first direction from the first end to the first pulley and in a second direction from the base member to the second end, wherein the second direction is different from the first direction.
4. The passenger seat of claim 3, wherein the first direction is perpendicular to the second direction.
5. The passenger seat of claim 1, wherein the second end of the cable is movable between a stowed position and a deployed position, and wherein the first pulley is linearly movable between a stowed position and a deployed position, and wherein a length of travel of the second end of the cable between the stowed position and the deployed position is different from a length of travel of the first pulley from the stowed position to the deployed position.
6. The passenger seat of claim 5, wherein the length of travel of the second end between the stowed position and the deployed position is twice the length of travel of the first pulley between the stowed position and the deployed position.
7. The passenger seat of claim 1, wherein the armrest control mechanism applies a resistive force to the armrest as the armrest moves in a downward direction, and the armrest control mechanism applies an assisting force to the armrest as the armrest moves in an upward direction.
8. The passenger seat of claim 1, further comprising a release latch attached to the armrest, wherein the release latch is operable between an engaged state and a disengaged state, wherein, in the engaged state, the release latch locks the armrest relative to the seat frame, and wherein, in the disengaged state, the armrest is moveable relative to the seat frame.
9. A method of assembling an armrest for a passenger seat relative to a seat frame of the passenger seat, the method comprising: supporting the armrest relative to the seat frame such that the armrest is vertically movable between a stowed position and a deployed position; and attaching an armrest control mechanism to the seat frame and the armrest for controlling vertical movement of the armrest relative to the seat frame, wherein the armrest control mechanism comprises: a base member; a gas spring comprising a first end attached to the base member and a second end moveable relative to the first end and relative to the base member; a first pulley attached to the second end of the gas spring and moveable relative to the base member; and a cable comprising a first end, a second end, and an intermediate portion between the first end and the second end, wherein the first end of the cable is attached to an anchor location on the base member, the second end of the cable is moveable relative to the base member, and the intermediate portion extends at least partially around and engages with the first pulley, wherein the first pulley is moveable based on a movement of the second end of the cable, and wherein attaching the armrest control mechanism to the seat frame and the armrest comprises attaching the base member to one of the seat frame or the armrest and attaching the second end of the cable to the other of the seat frame or the armrest.
10. The method of claim 9, wherein attaching the armrest control mechanism comprises guiding the cable such that the cable extends in a first direction from the first end to the first pulley and in a second direction from the base member to the second end, wherein the second direction is different from the first direction.
11. The method of claim 10, wherein the first direction is perpendicular to the second direction.
12. The method of claim 9, wherein the second end of the cable is movable between a stowed position and a deployed position, and wherein the first pulley is linearly movable between a stowed position and a deployed position, and wherein the method comprises guiding the cable on the armrest control mechanism such that a length of travel of the second end of the cable between the stowed position and the deployed position is different from a length of travel of the first pulley from the stowed position to the deployed position, wherein the length of travel of the second end

between the stowed position and the deployed position is twice the length of travel of the first pulley between the stowed position and the deployed position.

13. An armrest control mechanism comprising: a base member; a gas spring comprising a piston and cylinder, wherein the piston is movable relative to the cylinder, and wherein the cylinder is attached to the base member; a first pulley attached to the piston of the gas spring such that the first pulley is moveable relative to the base member; and a cable comprising a first end, a second end, and an intermediate portion between the first end and the second end, wherein the first end of the cable is attached to the base member, the second end of the cable is moveable relative to the base member, and the intermediate portion extends at least partially around and engages with the first pulley, wherein the first pulley is moveable based on a movement of the second end of the cable, and wherein the base member is attachable to one of a seat frame of a passenger seat or an armrest of the passenger seat and the second end of the cable is attachable to the other of the seat frame or the armrest.

14. The armrest control mechanism of claim 13, wherein the base member comprises a guide track, wherein the first pulley is attached to a guide member, and wherein the guide member is moveable along the guide track.

15. The armrest control mechanism of claim 13, further comprising a second pulley attached to the base member, wherein a guide path for the cable extends from an anchor location, around the first pulley, and around the second pulley toward the seat frame or the armrest.

16. The armrest control mechanism of claim 15, wherein the guide path for the cable between the anchor location and the first pulley is parallel to the guide path for the cable between the anchor location and the second pulley.

17. The armrest control mechanism of claim 15, wherein the guide path for the cable between the anchor location and the first pulley is in a first direction, and wherein a guide path for the cable from the second pulley toward the seat frame or the armrest is in a second direction different from the first direction.

18. The armrest control mechanism of claim 17, wherein the first direction is perpendicular to the second direction.

19. The armrest control mechanism of claim 18, wherein the gas spring and second pulley are supported on a first side of the base member, wherein the armrest further comprises a cable stopper for the cable on a second side of the base member, and wherein the guide path for the cable further extends from the first side of the base member to the second side of the base member.

20. The armrest control mechanism of claim 13, wherein the second end of the cable is movable between a stowed position and a deployed position, wherein the first pulley is linearly movable between a stowed position and a deployed position, wherein the cable is configured to be guided on the armrest control mechanism such that a length of travel of the second end of the cable between the stowed position and the deployed position is different from a length of travel of the first pulley from the stowed position to the deployed position, and wherein the length of travel of the second end between the stowed position and the deployed position is twice the length of travel of the first pulley between the stowed position and the deployed position.